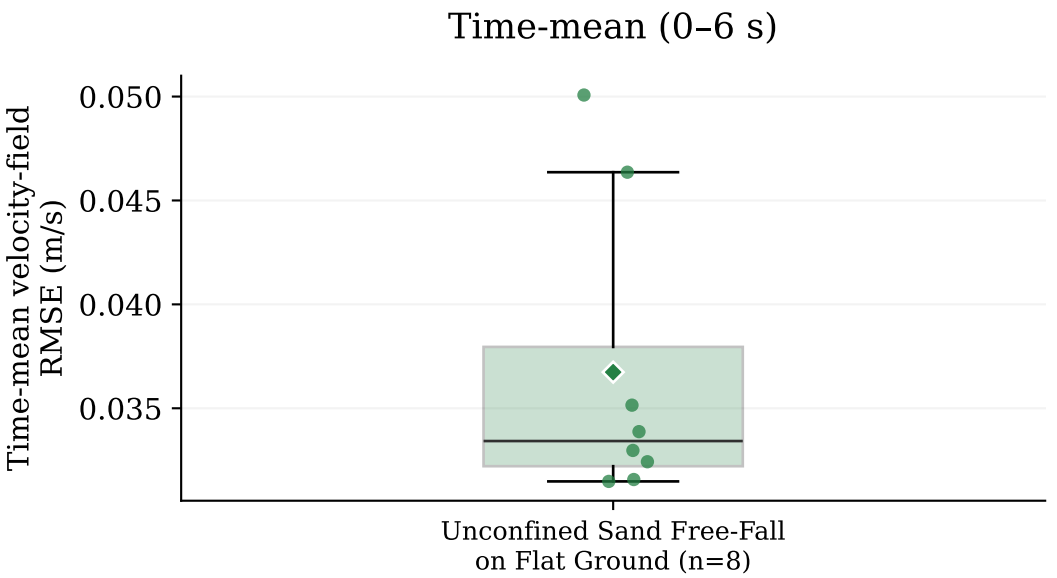
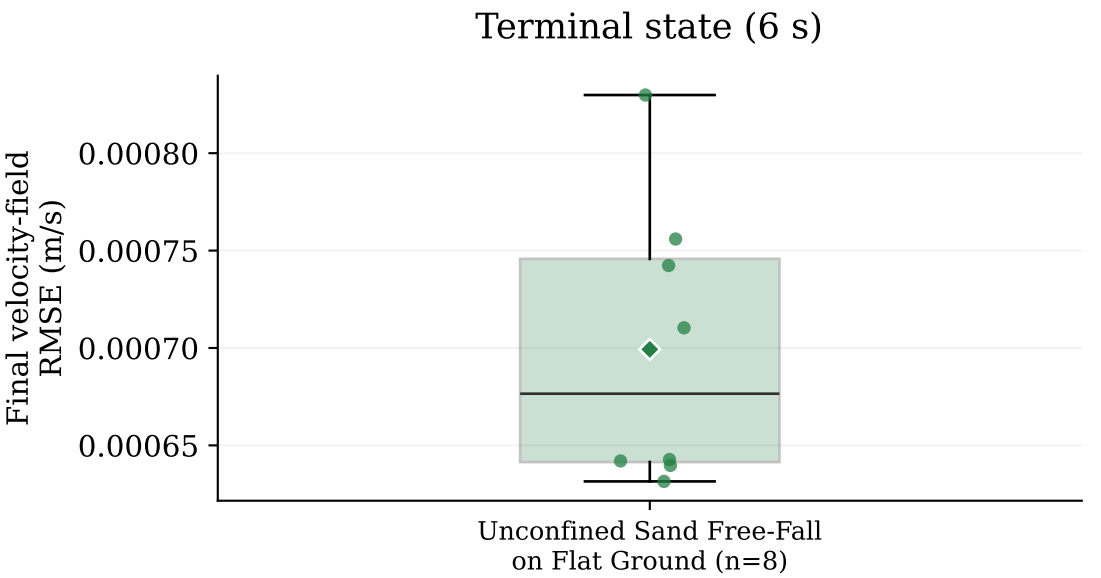
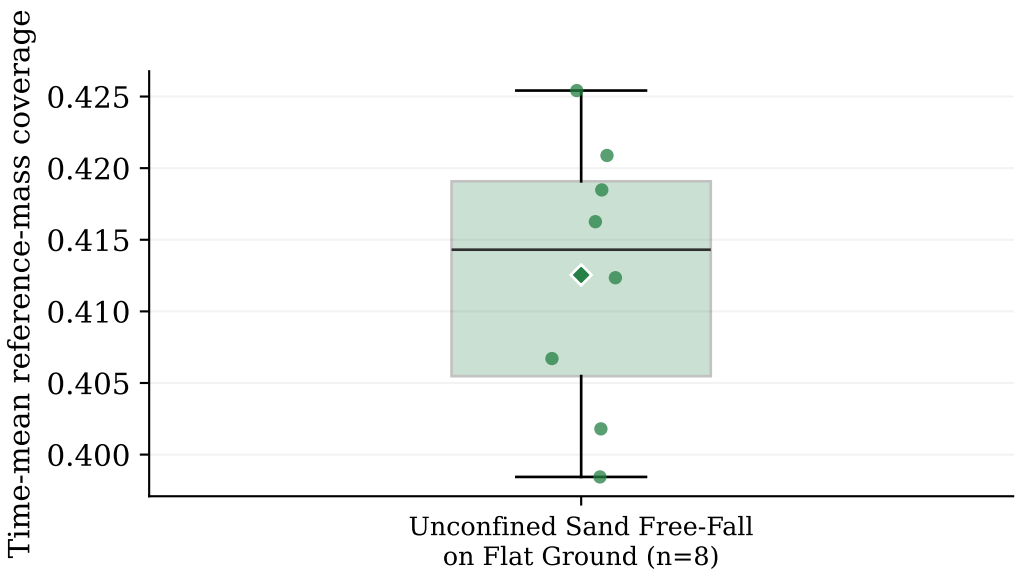
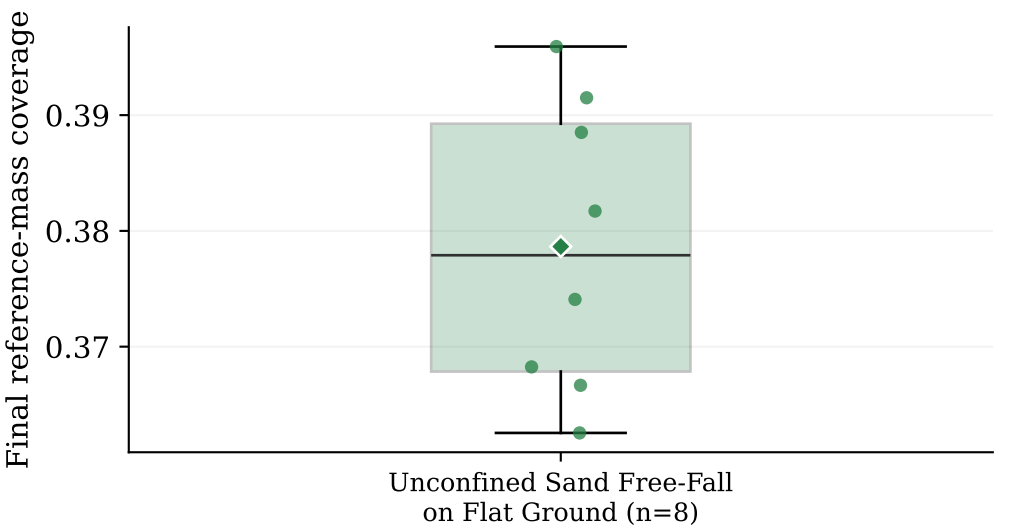
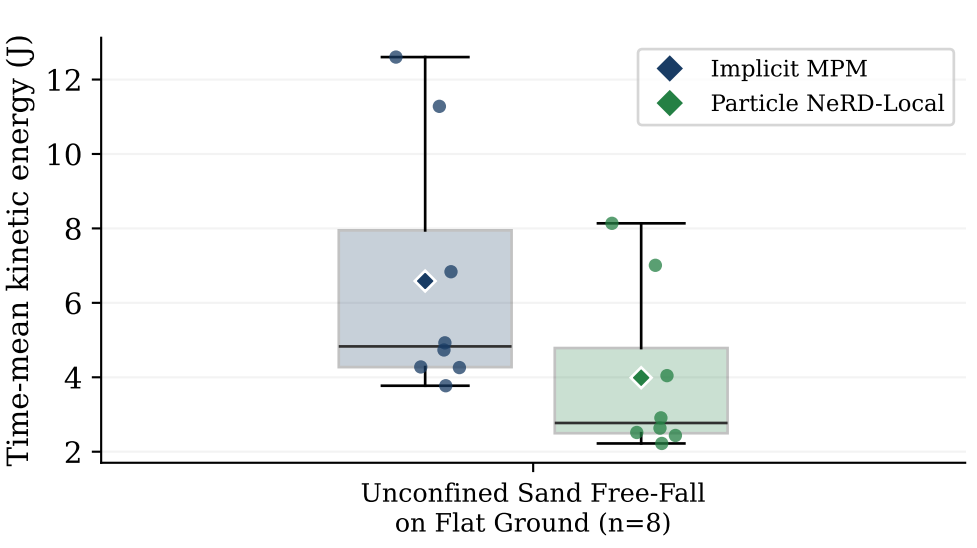
Group C: Unconfined Sand Free-Fall Motion on Flat Ground (No Rigid Object, n=8)











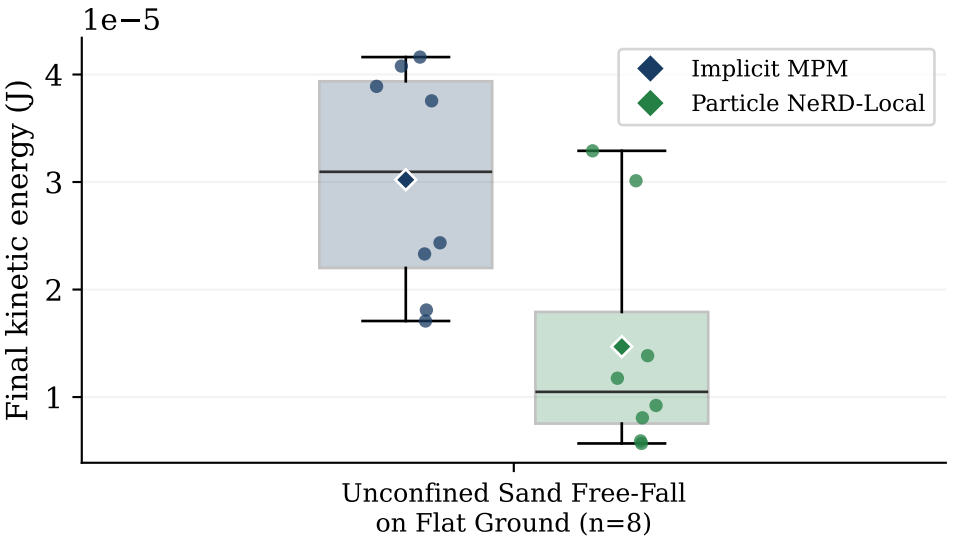


Figure 4c: Frozen object-model checkpoint 10,800. Velocity-field RMSE, reference-mass coverage, and particle kinetic energy for unconfined 3D sand blocks/columns dropped onto open flat ground (n=8).